

Data Analyst – Internship

Project: 1 – Financial KPI Analysis for a Startup

After project completion, prepare a 1–2 page report in PDF format, containing:

Introduction

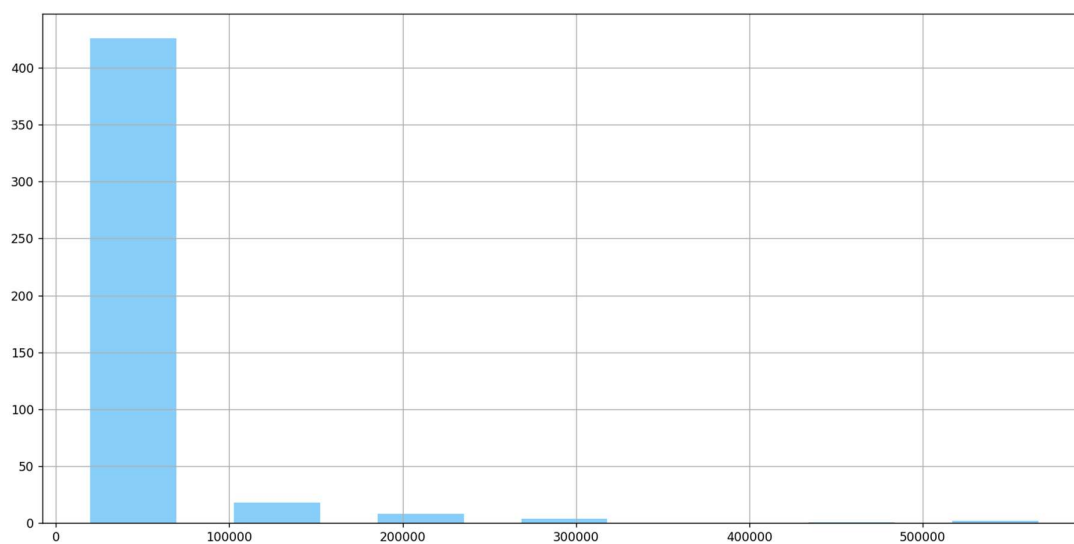
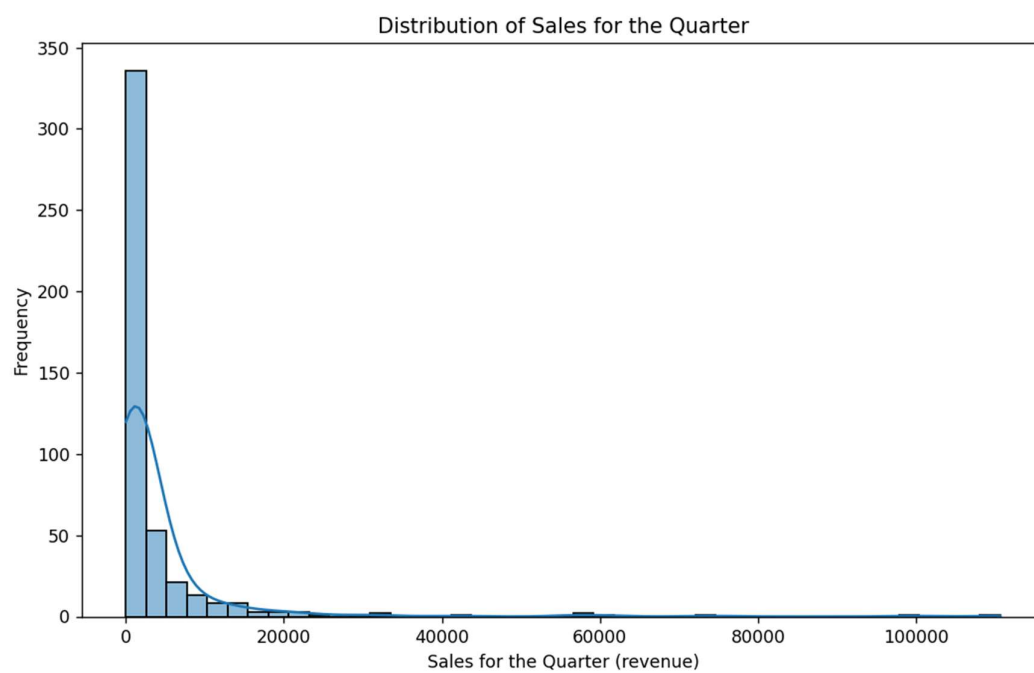
Abstract

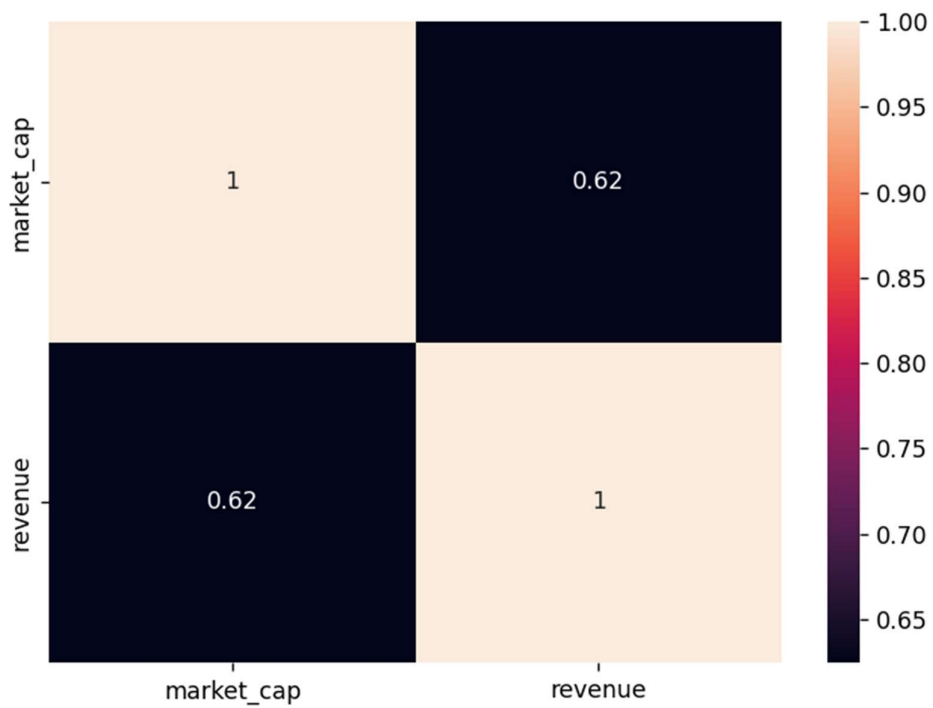
Tools Used

Steps Involved in Building the Project

Conclusion

Note: Report must not exceed 2 pages.





```
In [19]: pd.crosstab(df.market_cap, df.revenue)
Out[19]:
revenue    0.00    19.42    47.02    ...    74156.07    99810.00    110666.93
market_cap
3017.07      0      0      0    ...      0      0      0
3024.32      0      0      0    ...      0      0      0
3026.26      0      0      0    ...      0      0      0
3029.57      0      0      0    ...      0      0      0
3031.50      0      0      0    ...      0      0      0
...          ...      ...      ...    ...      ...      ...      ...
289497.37    0      0      0    ...      0      0      0
320985.27    0      0      0    ...      0      0      0
482953.59    0      0      0    ...      0      0      0
563709.84    0      0      0    ...      0      0      0
583436.72    0      0      0    ...      0      1      0

[459 rows x 459 columns]
```

```
In [21]: df.head
Out[21]:
<bound method NDFrame.head of
      Name market_cap ... customers_acquired total_customers
0 Reliance Inds. 583436.72 ... 998100 998100
1 TCS 563709.84 ... 309040 1307140
2 HDFC Bank 482953.59 ... 205812 1512952
3 ITC 320985.27 ... 97720 1610672
4 H D F C 289497.37 ... 168405 1779077
.. ... ... ...
454 Prime Focus 3031.50 ... 6096 17433647
455 Lak. Vilas Bank 3029.57 ... 7901 17441548
456 NOCIL 3026.26 ... 2492 17444040
457 Orient Cement 3024.32 ... 5115 17449155
458 Natl.Fertilizer 3017.07 ... 28407 17477562

[459 rows x 8 columns]>

In [22]: |
```

```
In [22]: X = df.iloc[ : , :-1]
        ...: X.head()
Out[22]:
      Name market_cap ... expenses customers_acquired
0 Reliance Inds. 583436.72 ... 69867.000 998100
1 TCS 563709.84 ... 21632.800 309040
2 HDFC Bank 482953.59 ... 14406.889 205812
3 ITC 320985.27 ... 6840.414 97720
4 H D F C 289497.37 ... 11788.357 168405

[5 rows x 7 columns]

In [23]:
```

```
[5 rows x 7 columns]
In [23]: y = df.iloc[:, -1]
        ...: y.head()
Out[23]:
0    998100
1    1307140
2    1512952
3    1610672
4    1779077
Name: total_customers, dtype: int64
```

```
In [25]: X = pd.get_dummies(X, columns=['revenue'])
...: X.head()
Out[25]:
```

	Name	market_cap	...	revenue_99810.0	revenue_110666.93
0	Reliance Inds.	583436.72	...	True	False
1	TCS	563709.84	...	False	False
2	HDFC Bank	482953.59	...	False	False
3	ITC	320985.27	...	False	False
4	H D F C	289497.37	...	False	False

[5 rows x 465 columns]

```
In [33]: df["burn_rate"] = df["expenses"] - df["revenue"]
In [34]: df
Out[34]:
```

	Name	market_cap	...	total_customers	burn_rate
0	Reliance Inds.	583436.72	...	998100	-29943.000
1	TCS	563709.84	...	1307140	-9271.200
2	HDFC Bank	482953.59	...	1512952	-6174.381
3	ITC	320985.27	...	1610672	-2931.606
4	H D F C	289497.37	...	1779077	-5052.153
...	...	...	...	...	...
454	Prime Focus	3031.50	...	17433647	-182.883
455	Lak. Vilas Bank	3029.57	...	17441548	-237.051
456	NOCIL	3026.26	...	17444040	-74.781
457	Orient Cement	3024.32	...	17449155	-153.459
458	Natl.Fertilizer	3017.07	...	17477562	-852.225

[459 rows x 9 columns]

```
In [37]: df["ltv"] = (df["revenue"] / df["total_customers"].replace(0, np.nan)) * 12
In [38]: df
Out[38]:
```

	Name	market_cap	revenue	...	burn_rate	cac	ltv
0	Reliance Inds.	583436.72	99810.00	...	-29943.000	0.070000	1.200000
1	TCS	563709.84	30904.00	...	-9271.200	0.070000	0.283709
2	HDFC Bank	482953.59	20581.27	...	-6174.381	0.070000	0.163241
3	ITC	320985.27	9772.02	...	-2931.606	0.070000	0.072805
4	H D F C	289497.37	16840.51	...	-5052.153	0.070000	0.113590
...	...	...	...	...	...	...	...
454	Prime Focus	3031.50	609.61	...	-182.883	0.070001	0.000420
455	Lak. Vilas Bank	3029.57	790.17	...	-237.051	0.070006	0.000544
456	NOCIL	3026.26	249.27	...	-74.781	0.070020	0.000171
457	Orient Cement	3024.32	511.53	...	-153.459	0.070004	0.000352
458	Natl.Fertilizer	3017.07	2840.75	...	-852.225	0.070001	0.001950

[459 rows x 11 columns]

```
[459 rows x 9 columns]
In [35]: df["cac"] = df["expenses"] / df["customers_acquired"].replace(0, np.nan)
In [36]: df
Out[36]:
```

	Name	market_cap	...	burn_rate	cac
0	Reliance Inds.	583436.72	...	-29943.000	0.070000
1	TCS	563709.84	...	-9271.200	0.070000
2	HDFC Bank	482953.59	...	-6174.381	0.070000
3	ITC	320985.27	...	-2931.606	0.070000
4	H D F C	289497.37	...	-5052.153	0.070000
...	...	...	...	...	...
454	Prime Focus	3031.50	...	-182.883	0.070001
455	Lak. Vilas Bank	3029.57	...	-237.051	0.070006
456	NOCIL	3026.26	...	-74.781	0.070020
457	Orient Cement	3024.32	...	-153.459	0.070004
458	Natl.Fertilizer	3017.07	...	-852.225	0.070001

```
In [39]: df["ltv_cac_ratio"] = df["ltv"] / df["cac"]
In [40]: df
Out[40]:
```

	Name	market_cap	revenue	...	cac	ltv	ltv_cac_ratio
0	Reliance Inds.	583436.72	99810.00	...	0.070000	1.200000	17.142857
1	TCS	563709.84	30904.00	...	0.070000	0.283709	4.052992
2	HDFC Bank	482953.59	20581.27	...	0.070000	0.163241	2.332001
3	ITC	320985.27	9772.02	...	0.070000	0.072805	1.040063
4	H D F C	289497.37	16840.51	...	0.070000	0.113590	1.622719
...	...	...	...	...	...	...	...
454	Prime Focus	3031.50	609.61	...	0.070001	0.000420	0.005994
455	Lak. Vilas Bank	3029.57	790.17	...	0.070006	0.000544	0.007766
456	NOCIL	3026.26	249.27	...	0.070020	0.000171	0.002449
457	Orient Cement	3024.32	511.53	...	0.070004	0.000352	0.005025
458	Natl.Fertilizer	3017.07	2840.75	...	0.070001	0.001950	0.027863

[459 rows x 12 columns]

```
In [41]: df["run_rate"] = df["revenue"] * 4
In [42]: df
Out[42]:
```

	Name	market_cap	revenue	...	ltv	ltv_cac_ratio	run_rate
0	Reliance Inds.	583436.72	99810.00	...	1.200000	17.142857	399240.00
1	TCS	563709.84	30904.00	...	0.283709	4.052992	123616.00
2	HDFC Bank	482953.59	20581.27	...	0.163241	2.332001	82325.08
3	ITC	320985.27	9772.02	...	0.072805	1.040063	39088.08
4	H D F C	289497.37	16840.51	...	0.113590	1.622719	67362.04
...	...	...	...	...	...	...	...
454	Prime Focus	3031.50	609.61	...	0.000420	0.005994	2438.44
455	Lak. Vilas Bank	3029.57	790.17	...	0.000544	0.007766	3160.68
456	NOCIL	3026.26	249.27	...	0.000171	0.002449	997.08
457	Orient Cement	3024.32	511.53	...	0.000352	0.005025	2046.12
458	Natl.Fertilizer	3017.07	2840.75	...	0.001950	0.027863	11363.00

[459 rows x 13 columns]

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In [43]: summary = {
...:     "Average Revenue": round(df["revenue"].mean(), 2),
...:     "Average Burn Rate": round(df["burn_rate"].mean(), 2),
...:     "Average CAC": round(df["cac"].mean(), 2),
...:     "Average LTV": round(df["ltv"].mean(), 2),
...:     "Average LTV:CAC Ratio": round(df["ltv_cac_ratio"].mean(), 2),
...:     "Current Annualized Run Rate": round(df["run_rate"].iloc[-1], 2)
...: }
In [44]: print("==== KPI SUMMARY =====")
...: for k, v in summary.items():
...:     print(f"{k}: {v}")
...:
==== KPI SUMMARY =====
Average Revenue: 3807.79
Average Burn Rate: -1142.34
Average CAC: 0.07
Average LTV: 0.01
Average LTV:CAC Ratio: 0.14
Current Annualized Run Rate: 11363.0

```